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Methodological Project Report

Selective Inference for Hierarchical Clustering

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1 Introduction

Introduction text...

2 Classical versus selective testing

2.1 Hypothesis testing and *p-values*

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $(E, \mathcal{E})$ a topological space and $\mathcal{T}$ the σ-algebra generated by $\mathcal{E}$. A *random variable* is a measurable function $X : (\Omega, \mathcal{F}) \to (E, \mathcal{T})$. The *(probability) distribution* of X is the mapping $P : \mathcal{T} \to [0, 1]$ such that $P(O) = (\mathbb{P} \circ X^{-1})(O)$ for all $O \in \mathcal{T}$. We say that P is *supported* on E and denote by $\mathcal{M}_1(E)$ the set of all probability distributions supported on $\mathcal{E}$. From now on, we will set $E = \mathbb{R}$ and simply write $\mathcal{M}(\mathbb{R}) = \mathcal{M}$.

Definition 1 (Hypothesis, Definition 1.1 in [4])**.** *A hypothesis $\mathcal{H}_0$ is a set of probability distributions in $\mathcal{M}$. A hypothesis is simple if it is a singleton, such as $\{P\}$ or $\{Q\}$, and composite otherwise. The complementary set $\mathcal{M} \setminus \mathcal{H}_0$ is called the alternative hypothesis of $\mathcal{H}_0$.*

Definition 2 (Test)**.** *A test for $\mathcal{H}_0$ is a binary partition of the sample space, defined by a mapping:*

$$\begin{array}{rcl} \pi_{\mathcal{H}_0} : \Omega & \to & \{0, 1\} \\ \omega & \mapsto & \pi_{\mathcal{H}_0}(\omega). \end{array} \tag{1}$$

For $\omega \in \Omega$, we say that the test rejects $\mathcal{H}_0$ based on ω if $\pi_{\mathcal{H}_0}(\omega) = 1$, and does not reject $\mathcal{H}_0$ based on ω if $\pi_{\mathcal{H}_0}(\omega) = 0$.

In what follows, we will adopt the more standard notation:

$$\mathbb{P}\left(\text{Reject } \mathcal{H}_0\right) = \mathbb{P}\left(\{\omega \in \Omega \,:\, \pi_{\mathcal{H}_0}(\omega) = 1\}\right),$$

making the dependence on π implicit.

Definition 3 (Type I error)**.** *We say that a test controls the type I error at level α if*

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) \leq \alpha, \quad \textit{for } \alpha \in (0, 1).$$

We say that it controls the type I error exactly at level α if

$$\mathbb{P}(\text{Reject } \mathcal{H}_0) = \alpha, \quad \textit{for } \alpha \in (0, 1).$$

Definition 4 (Stochastic dominance)**.** *Let X and Y be real-valued random variables. We say that Y stochastically dominates X, and write*

$$X \preceq_{\text{st}} Y,$$

if

$$\mathbb{P}(X \leq u) \geq \mathbb{P}(Y \leq u), \quad \forall\, u \in \mathbb{R}. \tag{2}$$

Definition 5 (Super-uniform random variable)**.** *Let X be a real-valued random variable. We say that X is super-uniform on [0,1], and write $X \sim \mathcal{SU}(0,1)$, if X stochastically dominates a uniform random variable on $[0,1]$, that is, if*

$$\mathbb{P}(X \leq u) \leq u, \quad \forall\, u \in [0,1]. \tag{3}$$

Proposition 1. *Let X and Y be real-valued random variables, with cumulative distribution functions F_X and F_Y, respectively. If $X \preceq_{\text{st}} Y$, then $F_X(Y) \sim \mathcal{SU}(0,1)$.*

Remark 1. *If $X = Y$, then $F_X(X)$ is super-uniform on $[0,1]$. Moreover, if X has a continuous distribution function F_X, then*

$$F_X(X) \sim \mathcal{U}(0,1). \tag{4}$$

Definition 6 (p-value, Definition 1.1 in [4])**.** *Let $\mathcal{H}_0$ be a hypothesis. A p-value for $\mathcal{H}_0$ is a super-uniform random variable under $\mathcal{H}_0$.*

We often build tests using p-values as $\pi = \mathbb{1}\{p \leq \alpha\}$.

Proposition 2. *Let $\mathcal{H}_0$ be a hypothesis and p a p-value for $\mathcal{H}_0$. Then, the test $\pi = \mathbb{1}\{p \leq \alpha\}$ controls the type I error at level α, for $\alpha \in (0,1)$.*

Definition 7 (Test statistic)**.** *A test statistic is a measurable function $T : \Omega \to \mathbb{R}$.*

From now on, we will consider the case of unilateral tests. In this setting, to define a test based on the information given by a real-valued random variable X, p-values are often built in the form:

$$p(x) = \mathbb{P}_{\mathcal{H}_\prime}(T(X) \geq T(x)) \tag{5}$$

where T is a test statistic and x a realization of X.

The p-value (5) follows a uniform distribution under $\mathcal{H}_0$. However, we have defined p-values as being super-uniform random variables under the null hypothesis, which corresponds to a more general family of distributions. To adapt the classical form (5) to that setting, we introduce the following proposition.

Proposition 3. *Let T and T' be two test statistics. Let X be a random variable and x a realization of X. Define*

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \geq T(x)\big). \tag{6}$$

If

$$T'(X) \preceq_{\text{st}} T(X) \quad \textit{under } \mathcal{H}_0, \tag{7}$$

then p is a p-value for $\mathcal{H}_0$.

Remark 2. *If $T' = T$ and the cumulative distribution function F_T of $T(X)$ is continuous under $\mathcal{H}_0$, then*

$$p \overset{\mathcal{H}_0}{\sim} \mathcal{U}(0,1).$$

In the classical setting, the null hypothesis is independent of the data. However, in many practical applications $\mathcal{H}_0$ is chosen *after seeing the data*. In this framework, the classical testing approaches built for type I error control are unsuitable. Instead, statistical guarantees need to be provided via *selective inference*. In particular, the theory of statistical testing of data-driven null hypotheses is known as *selective testing* [2]. The next section provides some examples that motivate this framework.

2.2 Examples motivating selective testing

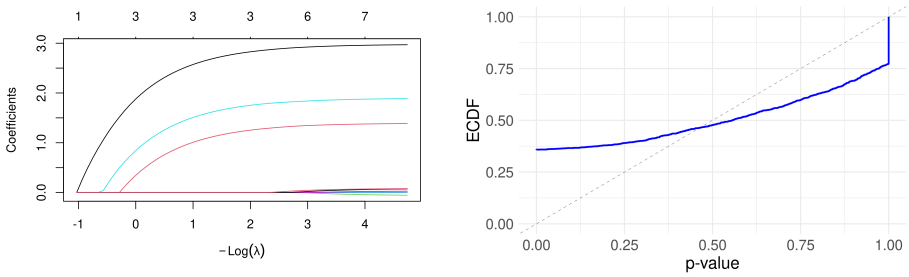
2.2.1 Lasso

To determine whether an explanatory variable helps explain a response variable through a linear regression model, a common practice is to test whether its associated coefficient is significantly different from zero. In the context of classical linear regression, this falls within the framework of non-selective inference, because all coeficients are fixed *a priori*. In contrast, Lasso regression uses an ℓ_1-penalty to perform variable selection [5]. As only the coefficients selected by the Lasso can be tested, the null hypothesis depends on the outcome of the Lasso regression, and is therefore data-driven. This corresponds to a setting of selective inference, where the classical control of type I error fails.

To illustrate the unsuitability of non-selective inference, consider a centered Gaussian vector $X = (X_1, \ldots, X_8)$ and set the model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1), \quad \beta = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top.$$

We generated $n = 100$ independent realizations of (X, Y) and implemented the Lasso algorithm, obtaining the regularization path presented in Figure 1(a). Then, we tested whether a randomly chosen coefficient selected by Lasso regression and among the trully null coefficients, equals 0. After repeating this pipeline $M = 2000$ times, we obtained the empirical p-value distribution depicted in Figure 1(b). We clearly see that the p-values are not super-uniform, motivating the use of a selective testing approach instead of naive testing after selection.



(a) (b)

Figure 1: (a) Regularization path of the coefficients obtained using the Lasso algorithm with a single realization of (X, Y). Some coefficients are zero, and three converge toward the theoretical values (3,1.5,2). (b) Empirical cumulative distribution function (ECDF) of the p-values obtained from tesing wheter a randomly chosen coefficient is null after selecting the coefficient by a Lasso regression. The ECDF was computed from $M = 2000$ simulations.

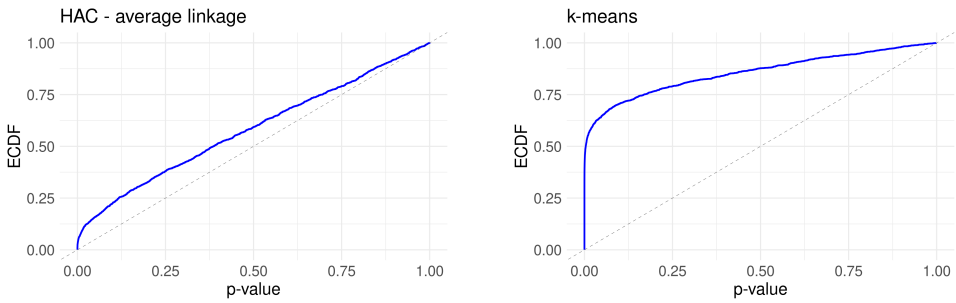
2.2.2 Publication bias

In many areas of research, scientists tend to test for signficance only when the associated effect is found to be substantial. In other words, testing is performed after a selection process that filters out small effects. In that setting, controlling type I error at a fixed level α yields inflated false positive rates when considering the ensemble of all published studies. If $Y_i \sim \mathcal{N}(\mu_i, 1)$ represents the effect size in a scientific study, whose significance is tested only if $|Y_i| > 1$, a naive level α test $H_{0,i} : \mu_i = 0$ is invalid. Indeed, Fithian [2] demonstrates that the false positive rate among true nulls reaches approximately 0.16, far exceeding the nominal 0.05 level. Valid inference requires thresholding $|Y_i|$ at 2.41 rather than 1.96, the 0.95 quantile of the standard normal, imposing a more stringent criterion.

2.2.3 Clustering

Another remarkable example of selective inference appears when evaluating the performance of clustering algorithms by testing for the equality of cluster means. To illustrate the need of using appropriate tests in the context of selective inference, we simulated $n = 100$ samples of a five-dimensional standard Gaussian random vector. Each sample was classified into $K = 3$ groups using hierarchical agglomerative clustering and k-means. Then, the equality of cluster means was tested using a classical t-test, for two randomly selected clusters. If the test controlled the type I error in this setting, the resulting p-value would be super-uniformly distributed under the null hypothesis. However, the 'clustering + post-selection testing' procedure leads

to a deviation from super-uniformity, as shown in Figure 2.



(a) (b)

Figure 2: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution.

2.3 Addressing selective testing

Since the type I error is no longer controlled, the definition of alternative error risks is required in this setting. Accordingly, we present three methods as described by Yoav Benjamini [1].

2.3.1 Simultaneous Inference

This approach controls the family-wise error rate across all hypotheses:

$$\mathbb{P}(\text{At least 1 false positive among all hypotheses}) \leq \alpha.$$

This strategy proves highly conservative, ensuring that for every possible set of hypotheses, the probability of at least one false positive remains below α. To date, no post-clustering inference approaches based on this type of control have been proposed.

2.3.2 Sample splitting

This approach consists in splitting the dataset into a training set X and a test set Y. The training set X is used to choose the null hypothesis $\mathcal{H}_\prime(X)$. Then, inference is performed on the test set Y.

Although relatively simple to implement, this method raises several issues. First, statistical guarantees on the tests hold only if X and Y are independent, which is rarely the case in practice. In addition, comparing cluster means on the observations in Y requires assigning each test point to one of the clusters obtained from the clustering performed on X, a step that

compromises validity, as it reuses the dataset X for two separate steps. As discussed in [3], this strategy does not yield valid post-clustering inference in general.

2.3.3 Conditional Inference

This approach constitutes the most extensively studied framework for post-clustering inference. It controls the false positive rate conditional on hypothesis selection:

$$\mathbb{P}(\text{Reject } \mathcal{H}_\prime(X) \mid \mathcal{H}_\prime \text{ selected}) \leq \alpha.$$

In the remainder of the article, we employ this method to develop a statistical procedure for testing equality of cluster means following a clustering algorithm.

3 Post-clustering inference

The article under study [3] was the first one to propose a feasible solution to post-clustering inference. Here, we will focus on the setting and approach introduced by the authors.

3.1 Gao *et al.*'s approach

Let $\mathbf{X} = (X_{ij})_{ij} \in \mathbb{R}^{n \times p}$ be the design matrix. A cluster is an element of a partition of the samples. We note C a clustering algorithm and $C_1 \in C(\mathbf{X})$ a cluster obtained by the algorithm C on $\mathbf{X}$.

We note $\boldsymbol{\mu} = (\mu_{ij})_{ij}$ where $\mu_{ij} = \mathbb{E}[X_{ij}]$. For a subset G of $\{1, \ldots, n\}$, we note $\bar{\mu}_G = \dfrac{1}{|G|} \displaystyle\sum_{i \in G} \mu_i \in \mathbb{R}^p$ and $\bar{X}_G = \dfrac{1}{|G|} \displaystyle\sum_{i \in G} X_i \in \mathbb{R}^p$.

The goal is to test for the equality of cluster means, that is, testing the following null hypothesis:

$$\mathcal{H}_0^{C_1, C_2} : \bar{\mu}_{C_1} = \bar{\mu}_{C_2} \tag{8}$$

Make sure your notation is consistent: here you have $\mathcal{H}_0^{C_1,C_2}$, but below you have H_0 and $\mathcal{H}_0^{\{C_1,C_2\}}$.. chose only one. For concision and clarity, we introduce the n-dimensional vector

$$\nu = (\nu_i)_{i=1}^n, \qquad \nu_i = \begin{cases} \dfrac{1}{|C_1|}, & \text{if } i \in C_1, \\ \dfrac{1}{|C_2|}, & \text{if } i \in C_2, \\ 0, & \text{otherwise,} \end{cases}$$

such that the population (resp. empirical) difference of cluster means can be compactely rewritten as:

$$\begin{cases} \bar{\mu}_{C_1} - \bar{\mu}_{C_2} = \nu^T \boldsymbol{\mu} \\ \bar{X}_{C_1} - \bar{X}_{C_2} = \nu^T \mathbf{X} \end{cases}$$

Thus, we can rewrite the null hypothesis as follows:

$$\mathcal{H}_\prime : \bar{\mu}_{C_1} = \bar{\mu}_{C_2} \iff \nu^T \boldsymbol{\mu} = 0. \tag{9}$$

Here, say that we are adopting the conditional inference framework described in the previous section which, in the setting of clustering, is equivalent to control the so-called selective type I error for clustering, defined below.

Definition 8 (Selective type I error for clustering)**.** *We say that a test controls the selective type I error for clustering at leval α if*

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}(\text{Reject } \mathcal{H}_0^{\{C_1,C_2\}} \mid C_1, C_2 \in C(\mathbf{X})) \leq \alpha, \quad \alpha \in (0,1).$$

We say that it controls exactly the selective type I error for clustering at level α if

$$\mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}(\text{Reject } \mathcal{H}_0^{\{C_1,C_2\}} \mid C_1, C_2 \in C(\mathbf{X})) = \alpha, \quad \alpha \in (0,1).$$

To defined a p-value that controls the selective type I error, model assumptions need to be imposed on $\mathbf{X}$. The authors in [3] adopt the following matrix normal model:

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \sigma^2 \mathbb{I}_p), \tag{10}$$

Don't indent after equations (unless you start a new paragraph). To avoid this, don't leave blank lines between a equation and the text. which means that the lines of $\mathbf{X}$ are independent p-dimensional random vectors distributed as $\mathcal{N}_p(\mu_i, \sigma^2 \mathbb{I}_p)$, for every line $i \in \{1, \ldots, n\}$. Note that this model imposes an independence assumption between the features (columns) of $\mathbf{X}$.

To run a proper test, we need to control this error what error? at level α. Ideally, we would like to define a p-value as follows:

$$p_{ideal}(x) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}(T(\mathbf{X}) \geq T(x) \mid C_1, C_2 \in C(\mathbf{X})),$$

with T being a test statistic. In [3], they defined the test statistic as $T(\mathbf{X}) = ||\nu^\top \mathbf{X}||_2$, because its distribution under the null hypothesis can be straightforwardly derived. Indeed, noting that.

$$\nu^\top \mathbf{X} \sim \mathcal{MN}_{1\times p}(\nu^\top \mu,\ \nu\nu^\top,\ \sigma^2 I_p) = \mathcal{N}_p\Big(\nu^\top \mu,\ \|\nu\|_2^2 \sigma^2 I_p\Big),$$

we have, under $\mathcal{H}_0^{\{C_1,C_2\}}$, that

$$||\nu^\top \mathbf{X}||_2 \sim ||\nu||_2 \sigma \chi_p. \tag{11}$$

Remove the first line of the previous equation: it is important to be more or less homogeneous in the level of intelligence that we assume the reader has.

However, p_{ideal} cannot be evaluated in practice as it depends on unknown parameters [3]. To address this issue, the authors in [3] propose to add technical events to the conditioning set, considering the following quantity:

$$p(x) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(||\nu^T \mathbf{X}||_2 \geq ||\nu^T x||_2 \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}), E(\mathbf{X})\right) \tag{12}$$

as a p-value for (9), with $E(\mathbf{X})$ being the additional events. To justify the choice of $E(\mathbf{X})$, we follow the steps in [3]. First, we decompose the data matrix as follows:

$$\mathbf{X} = \pi_\nu^\perp \mathbf{X} + (\mathbb{I}_n - \pi_\nu^\perp)\mathbf{X} \tag{13}$$

where $\pi_\nu^\perp$ is the orthogonal projection matrix onto the subspace orthogonal to ν, defined as:

$$\pi_\nu^\perp = \mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2} \in \mathbb{R}^{n\times n}. \tag{14}$$

Then, we can write

$$\begin{aligned}
\mathbf{X} &= \pi_\nu^\perp \mathbf{X} + \frac{\nu\nu^T}{||\nu||_2^2}\mathbf{X} \\
&= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\frac{\nu^T\mathbf{X}}{||\nu^T\mathbf{X}||_2}||\nu^T\mathbf{X}||_2 \\
&= \pi_\nu^\perp \mathbf{X} + \frac{\nu}{||\nu||_2^2}\,\mathrm{dir}(\nu^T\mathbf{X})||\nu^T\mathbf{X}||_2,
\end{aligned} \tag{15}$$

where dir is the direction of a vector defined as:

$$\mathrm{dir}(z) = \frac{z}{||z||_2}, z \in \mathbb{R}^p. \tag{16}$$

Since $\mathrm{dir}(\nu^T\mathbf{X})$ and $\pi_\nu^\perp\mathbf{X}$ are unknown, we condition on their values for a realization $\mathbf{x}$ of $\mathbf{X}$, defining the additional events as:

$$E(\mathbf{X}) = \{\pi_\nu^\perp \mathbf{X} = \pi_\nu^\perp \mathbf{x},\ \mathrm{dir}(\nu^T\mathbf{X}) = \mathrm{dir}(\nu^T\mathbf{x})\}$$

Use also bold notation for the realization $\mathbf{x}$ of $\mathbf{X}$. Plugging the previous equation in (12) we obtain the following expression for the p-value:

$$p(x) = \mathbb{P}_{\mathcal{H}_0^{\{C_1,C_2\}}}\left(||\nu^T\mathbf{X}||_2 \geq ||\nu^T x||_2 \mid C_1, C_2 \in \mathcal{C}(\mathbf{X}),\ \pi_\nu^\perp\mathbf{X} = \pi_\nu^\perp x,\ \mathrm{dir}(\nu^T\mathbf{X}) = \mathrm{dir}(\nu^T x)\right) \tag{17}$$

Separate the previous equation in two lines, with the additional events in the second one.

Now we haved defined the model, the hypothesis to test and the candidate p-value. The goal now is to prove that the candidate p-value (i) can be computed under an analitically tractable form and (ii) controls the selective type I error for clustering. Now, we say that to prove all that, we first need a technical lemma that we state now (the Lemma about the independence that we proved). Add the lemma here and its proof to the appendix.

Lemma 1. *Under the null hypothesis, the following statements hold:*

$$\begin{aligned} &(i) \quad \|\nu^\top \mathbf{X}\|_2 \perp\!\!\!\perp \pi_\nu^\perp \mathbf{X}, \\ &(ii) \quad \|\nu^\top \mathbf{X}\|_2 \perp\!\!\!\perp \mathrm{dir}(\nu^\top \mathbf{X}). \end{aligned}$$

Now, we can rewrite the p-value **??** as:

$$p(x) = ... \tag{18}$$

So far, we proved the first part of the theorem 1 in [3], the proof of the second part is in the annexes.

Theorem 1 (Theorem 1 in [3])**.** *For any realization* $\mathbf{x}$ *from* $\mathbf{X} \sim \mathcal{MN}_{n\times p}(\mu, \mathbb{I}_n, \sigma^2 \mathbb{I}_p)$*, and for any non-overlapping groups of observations* $C_1, C_2 \subseteq \{1, ..., n\}$*,*

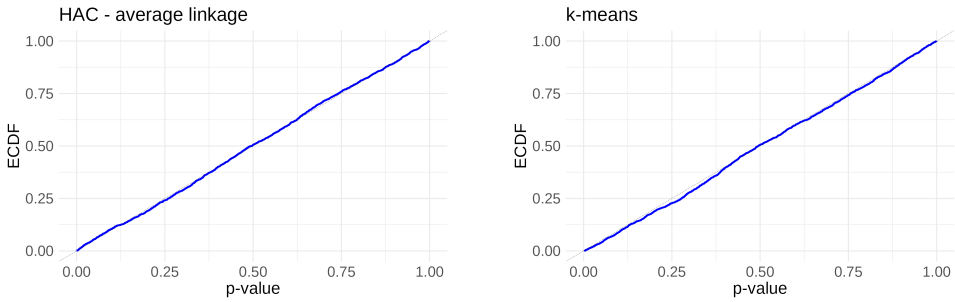
$$p(\mathbf{x}, \{C_1, C_2\}) = 1 - \mathbb{F}(, , \mathcal{S}(\mathbf{x}, \{\mathcal{C}_\infty, \mathcal{C}_\in\})) \tag{19}$$

4 Numerical experiments

Section 3.1 detailed the approach of Gao et al. for testing the equality of cluster means after applying a hierarchical agglomerative clustering (HAC) algorithm. Theorem **??** ensures that the p-value defined in (**??**) controls the selective type I error.

To empirically illustrate this result, we performed numerical simulations in the same setting as in Section 2.2.3, changing only the test statistic. For the simulations based on the HAC algorithm, we used the `clusterpval` library developed by Gao et al., which implements the p-value defined in (**??**). For the simulations based on the k-means algorithm, we used the `KmeansInference` library, which relies on a similar p-value adapted to the k-means procedure.

Figure 3 shows that the resulting p-values are indeed super-uniform.



(a) (b)

Figure 3: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution.

5 Discussion

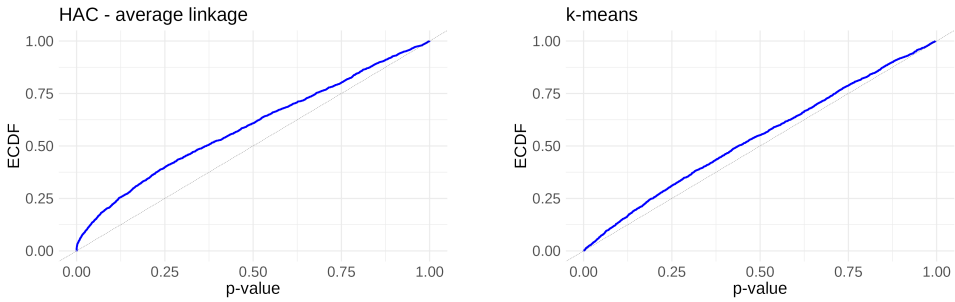
The control of the selective type I error for clustering relies on a strong assumption: the observations must be independent.

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbb{I}_n, \sigma^2 \mathbb{I}_p),$$

To determine whether the method is robust to this assumption, we simulated centered Gaussian random vector with correlation matrix:

$$R = 0.5^{|i-j|} \text{ for } 1 \leq i, j \leq p.$$

Figure 4 shows that the p-values are no longer super-uniform, and therefore the selective type I error is no longer controlled.



(a) (b)

Figure 4: Empirical cumulative distribution functions of the p-values obtained after testing the equality of cluster means for (a) hierarchical agglomerative clustering (HAC) and (b) k-means algorithms. The distribution functions were computed using $M = 2000$ simulations from a multivariate centered normal distribution.

A Proofs

A.1 Proofs of Section 2

Proof of Proposition 1. Let G_X denote the generalized inverse (quantile function) of F_X, defined for $u \in [0, 1]$ as

$$G_X(u) = \inf\{x \in \mathbb{R} : F_X(x) \geq u\}. \tag{20}$$

By definition of the generalized inverse,

$$\{F_X(Y) \leq u\} = \{Y < G_X(u)\}. \tag{21}$$

Therefore,

$$\mathbb{P}\big(F_X(Y) \leq u\big) = \mathbb{P}\big(Y < G_X(u)\big) = F_Y\big(G_X(u)^-\big), \tag{22}$$

where $G_X(u)^-$ denotes the left limit at $G_X(u)$.

Since $X \preceq_{\text{st}} Y$, we have $F_Y \leq F_X$ pointwise, and thus

$$F_Y\big(G_X(u)^-\big) \leq F_X\big(G_X(u)^-\big). \tag{23}$$

By the defining property of the generalized inverse,

$$F_X\big(G_X(u)^-\big) \leq u. \tag{24}$$

Combining these inequalities yields

$$\mathbb{P}\big(F_X(Y) \leq u\big) \leq u, \quad \forall\, u \in [0, 1], \tag{25}$$

which proves that $F_X(Y)$ is super-uniform.

□

Proof of Remark 1. By the proposition 1, we have that $F_X(X) \sim \mathcal{SU}(0,1)$. If F_X is continuous, then $F_X(G_X(u)) = u$ for all $u \in [0,1]$, and hence

$$\mathbb{P}\big(F_X(X) \leq u\big) = F_X\big(G_X(u)\big) = u, \quad \forall\, u \in [0,1], \tag{26}$$

which concludes the proof.

□

Proof of Proposition 2. By definition of the rejection rule,

$$\mathbb{P}_{\mathcal{H}_0}(\text{Reject } \mathcal{H}_0) = \mathbb{P}_{\mathcal{H}_0}(p \leq \alpha). \tag{27}$$

Since p is a p-value for $\mathcal{H}_0$, it is super-uniform under $\mathcal{H}_0$, hence

$$\mathbb{P}_{\mathcal{H}_0}(p \leq \alpha) \leq \alpha. \tag{28}$$

This establishes control of the type I error at level α. □

Proof of Proposition 3. Let $F_{T'(X)}$ denote the distribution function of $T'(X)$ under $\mathcal{H}_0$. By Proposition 1, the stochastic dominance $T'(X) \preceq_{\text{st}} T(X)$ implies that

$$F_{T'(X)}\big(T(X)\big) \sim \mathcal{SU}(0,1).$$

By definition,

$$p(x) = \mathbb{P}_{\mathcal{H}_0}\big(T'(X) \geq T(x)\big) = 1 - F_{T'(X)}\big(T(x)\big). \tag{29}$$

Let $u \in [0,1]$. Then

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_{T'(X)}(T(X)) \leq u\big) && (30)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \geq 1 - u\big) && (31)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \leq 1 - u\big)\,. && (32)
\end{aligned}$$

Since $F_{T'(X)}(T(X))$ is super-uniform,

$$\mathbb{P}_{\mathcal{H}_0}\big(F_{T'(X)}(T(X)) \leq 1 - u\big) \leq 1 - u, \tag{33}$$

and therefore

$$\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) \leq u. \tag{34}$$

Thus, p is super-uniform under $\mathcal{H}_0$, and hence a p-value. □

Proof of Remark 2. When $T' = T$

$$p(X) = \mathbb{P}_{\mathcal{H}_0}\big(T(X') \geq T(X)\big) = 1 - F_T\big(T(X)\big), \tag{35}$$

By Remark 1, if F_T is continuous,

$$F_T\big(T(X)\big) \sim \mathcal{U}(0,1).$$

Consequently, for any $u \in [0,1]$,

$$\begin{aligned}
\mathbb{P}_{\mathcal{H}_0}\big(p(X) \leq u\big) &= \mathbb{P}_{\mathcal{H}_0}\big(1 - F_T(T(X)) \leq u\big) && (36)\\
&= \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \geq 1 - u\big) && (37)\\
&= 1 - \mathbb{P}_{\mathcal{H}_0}\big(F_T(T(X)) \leq 1 - u\big) && (38)\\
&= 1 - (1 - u) && (39)\\
&= u. && (40)
\end{aligned}$$

Thus p is uniformly distributed on $[0,1]$ under $\mathcal{H}_0$. □

A.2 Proofs of Section 3

Proof of Lemma 1. i)

$$\begin{aligned}
\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \mathbb{E}(\nu^\top \mathbf{X}\mathbf{X}^\top (\pi_\nu^\perp)^\top)\\
&= \nu^\top \mathbb{E}(\mathbf{X}\mathbf{X}^\top)(\pi_\nu^\perp)^\top\\
&= \nu^\top (\mathbb{V}(\mathbf{X}) + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top
\end{aligned} \tag{41}$$

Let $X \sim \mathcal{MN}(\mu, \mathbf{U}, \mathbf{\Sigma})$. Then, let us state $\mathbb{V}(\mathbf{X}) = \mathbf{U}\,\mathrm{Trace}(\mathbf{\Sigma})$. We prove the previous equality as a technical lemma after the proof.

In this case $\mathbb{V}(\mathbf{X}) = \mathbb{I}_n\,\mathrm{Trace}(\sigma^2 \mathbb{I}_p) = \sigma^2 p \mathbb{I}_n$

So

$$\begin{aligned}
\mathbb{E}(\nu^\top \mathbf{X}(\pi_\nu^\perp \mathbf{X})^T) &= \nu^\top (\sigma^2 p \mathbb{I}_n + \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top)(\pi_\nu^\perp)^\top\\
&= \sigma^2 p \nu^\top (\pi_\nu^\perp)^\top + \nu^\top \mathbb{E}(\mathbf{X})\mathbb{E}(\mathbf{X})^\top (\pi_\nu^\perp)^\top\\
&= \sigma^2 p \nu^\top (\mathbb{I}_n - \frac{\nu\nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}(\pi_\nu^\perp \mathbf{X})^\top\\
&= \sigma^2 p (\nu^\top - \frac{||\nu||_2^2 \nu^\top}{||\nu||_2^2}) + \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top)\\
&= \mathbb{E}(\nu^\top \mathbf{X})\mathbb{E}((\pi_\nu^\perp \mathbf{X})^\top)
\end{aligned} \tag{42}$$

ii)

$$\nu^T \mathbf{X} \sim \mathcal{MN}_{1\times p}(\nu^\top \mu, \nu\nu^\top, \sigma^2 \mathbb{I}_p) = \mathcal{N}_p(\nu^\top \mu, ||\nu||_2^2 \sigma^2 \mathbb{I}_p,) \tag{43}$$

So $\nu^T \mathbf{X} \sim \mathcal{N}_p(0, ||\nu||_2^2 \sigma^2 \mathbb{I}_p)$ under $\mathcal{H}_0$. Yet, the density of a centered Gaussian random vector depends only on its norm, due to symmetry with respect to the origin. So $||\nu^\top \mathbf{X}||$ and $\mathrm{dir}(\nu^\top \mathbf{X})$ are independent.

□

Lemma 2.

Proof of Lemma 2.

$$\begin{aligned} \text{Let } \mathbf{Y} = \mathbf{X} - \mu \\ \mathbb{V}(\mathbf{X}) &= \mathbb{E}[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^\top] \\ &= \mathbb{E}(\mathbf{Y}\mathbf{Y}^\top) \end{aligned} \tag{44}$$

And

$$\begin{aligned} (\mathbf{Y}\mathbf{Y}^T)_{ij} &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{kj}^\top \\ &= \sum_{k=1}^{p} \mathbf{Y}_{ik}\mathbf{Y}_{jk} \end{aligned} \tag{45}$$

Therefore,

$$\mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) = \sum_{k=1}^{p} \mathbb{E}(\mathbf{Y}_{ik}\mathbf{Y}_{jk}) \tag{46}$$

As $\mathbf{Y}$ is centered, by definition of a normal matrix we have:

$$\begin{aligned} \mathbb{E}((\mathbf{Y}\mathbf{Y}^\top)_{ij}) &= \sum_{k=1}^{p} \mathbf{U}_{ij}\Sigma_{kk} \\ &= \mathbf{U}_{ij} \sum_{k=1}^{p} \Sigma_{kk} \\ &= \mathbf{U}_{ij}\,\mathrm{Trace}(\Sigma) \end{aligned} \tag{47}$$

□

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